

WEEK 3 INTERNSHIP ASSIGNMENT

SQL & Excel for Data Analytics

Candidate Name:	Ayush
Course & Stream:	B.Tech in Computer Science & Engineering (IoT)
College / Institute:	Prestige Institute of Engineering, Management & Research, Indore
Assignment Title:	Week 3 — SQL & Excel for Data Analytics
Internship Track:	Data Analytics Internship
Technologies Practiced:	SQL (SQLite, Relational Queries), Microsoft Excel (openpyxl)
Assignment Marks:	100 Marks (Tasks 1 to 9 Complete)
Integrity Declaration:	Original student implementation; zero fabrication; verified execution.

Submission Note: This project represents genuine, authentic work completed for Week 3 of the Data Analytics internship. All SQL queries have been executed and verified against the SQLite database (company_analytics.db). All Excel functions, formatting rules, and chart models have been engineered into Week_3_Excel_Data_Analytics.xlsx. All calculations and outputs documented in this report are mathematically verified.

1. Introduction

In modern data analytics architectures, **Structured Query Language (SQL)** and **Microsoft Excel** represent the two most widely utilized and complementary analytical tools. While SQL serves as the industry-standard language for querying, joining, and transforming high-volume data stored in relational database management systems (RDBMS), Excel serves as the primary canvas for data presentation, business modeling, ad-hoc financial analysis, and visual communication.

Role of SQL in Data Analytics

SQL operates directly at the data warehouse and transactional database level. Data analysts use SQL to extract targeted slices from normalized schemas containing millions of records. Its declarative syntax allows analysts to express complex logic: filtering rows using WHERE, aggregating metrics with GROUP BY and HAVING, performing statistical calculations (COUNT, AVG, SUM, MIN, MAX), unifying multi-table entities via INNER, LEFT, and RIGHT JOIN, and nesting subqueries for multi-tiered filtering. SQL provides the reliable, structured data foundation.

Role of Microsoft Excel in Business Analytics

Once structured datasets are extracted, Microsoft Excel empowers business analysts to perform flexible tabular modeling. Excel provides intuitive cell formatting (currency, dates, borders, zebra striping), logical evaluation via IF(), conditional aggregations via COUNTIF() and SUMIF(), relational lookups via classic VLOOKUP() and modern XLOOKUP(), and multi-dimensional analysis through **Pivot Tables** and visual charts. Excel bridges the gap between raw database tables and executive decision-making.

Synergy in the Enterprise Analytics Pipeline

Professional data analysts rarely use SQL or Excel in isolation. Instead, an end-to-end analytics workflow follows a proven pipeline: (1) Ingest, clean, and join relational tables using SQL queries; (2) Export clean aggregates into Excel workbooks; (3) Apply conditional formatting and lookup formulas; and (4) Build interactive Pivot Tables and charts for stakeholders. Mastering both tools enables an analyst to move effortlessly from deep database querying to executive presentation.

What Was Practiced During Week 3

During this third week of the internship, practical exercises focused on progressive mastery:

- **Database Basics & Selection:** Database schemas, SELECT all, column projection, and column aliases.
- **Filtering & Sorting:** WHERE clauses, comparison operators, multi-column sorting (ORDER BY), and aggregate functions.
- **Group Aggregations:** GROUP BY on foreign keys, calculating group statistics, and group filtering with HAVING.
- **Relational Joins:** Demonstrating INNER JOIN, LEFT JOIN, and RIGHT JOIN on matching and unassigned records.
- **Subquery Engineering:** Single-row scalar subqueries, multi-row IN subqueries, and correlated subqueries.
- **Excel Data Hygiene:** Professional data formatting (currency, dates), zebra striping, descending sorts, and auto-filters.
- **Logical & Statistical Functions:** Implementing IF(), COUNTIF(), SUMIF(), and conditional formatting rules.
- **Relational Excel Lookups:** Exact match VLOOKUP() vs. modern bi-directional XLOOKUP() with error handling.
- **Pivot Tables & Data Visualization:** Pivot-style departmental summaries and Clustered Column budget charts.

2. Assignment Objectives

- 1. Understand database concepts and write SQL SELECT queries with column projection and aliases (Task 1).
- 2. Apply conditional filtering with WHERE, sort records with ORDER BY, and calculate aggregate functions (Task 2).
- 3. Construct GROUP BY queries to compute department metrics and filter grouped results using HAVING (Task 3).
- 4. Master relational joins by executing and explaining INNER JOIN, LEFT JOIN, and RIGHT JOIN (Task 4).
- 5. Implement nested SQL subqueries (scalar, multi-row, and correlated) to resolve multi-level analytical problems (Task 5).
- 6. Structure an Excel dataset with professional formatting, proper data types, sorting, and auto-filters (Task 6).
- 7. Implement Excel formulas using IF(), COUNTIF(), and SUMIF(), and configure conditional formatting rules (Task 7).
- 8. Perform cross-table lookups comparing classic VLOOKUP() with modern bi-directional XLOOKUP() (Task 8).
- 9. Construct a multi-dimensional Pivot Table summary and build an Excel Clustered Column Chart with commentary (Task 9).

3. Educational Dataset Documentation

Synthetic / Educational Data Notice: In strict compliance with internship integrity, all datasets used in this assignment are original educational synthetic datasets modeled after an enterprise Human Resources and Payroll Analytics system. The database is stored in Database/company_analytics.db and mirrored in Excel/Week_3_Excel_Data_Analytics.xlsx.

Table Name	Records	Attributes / Schema	Relational Role & Purpose
Departments	6 rows	Department_ID (PK), Department_Name, Manager_Name, Location, Budget	Parent catalog defining organizational divisions. Department 6 (R&D;) has zero assigned staff to demonstrate RIGHT JOIN.
Employees	25 rows	Emp_ID (PK), First_Name, Last_Name, Department_ID (FK), Job_Title, Salary, Hire_Date, Performance_Rating, City	Child transactional table with staff records. Employee 125 has a NULL Department_ID to demonstrate LEFT JOIN.

Relational Edge Cases Built for Testing: To provide rigorous demonstrations of SQL Joins in Task 4, the dataset contains: (1) **Unassigned Employee (Emp 125 Tanvi Bansal)** with Department_ID = NULL; and (2) **Unstaffed Department (Dept 6 Research & Development)** with a ₹1.1M budget but 0 assigned staff. This guarantees authentic mathematical distinctions between INNER, LEFT, and RIGHT joins.

Task 1: Introduction to Databases & SELECT Statement (10 Marks)

1. Objective: Understand core relational database concepts, query a sample employee table, retrieve all records using SELECT *, project specific columns, and apply column aliases and calculated fields.

2. Concepts Used: Relational tables, Primary Keys, SELECT, FROM, String concatenation (||), Column Aliases (AS), Arithmetic expressions.

3. SQL Source Queries (task_1.sql):

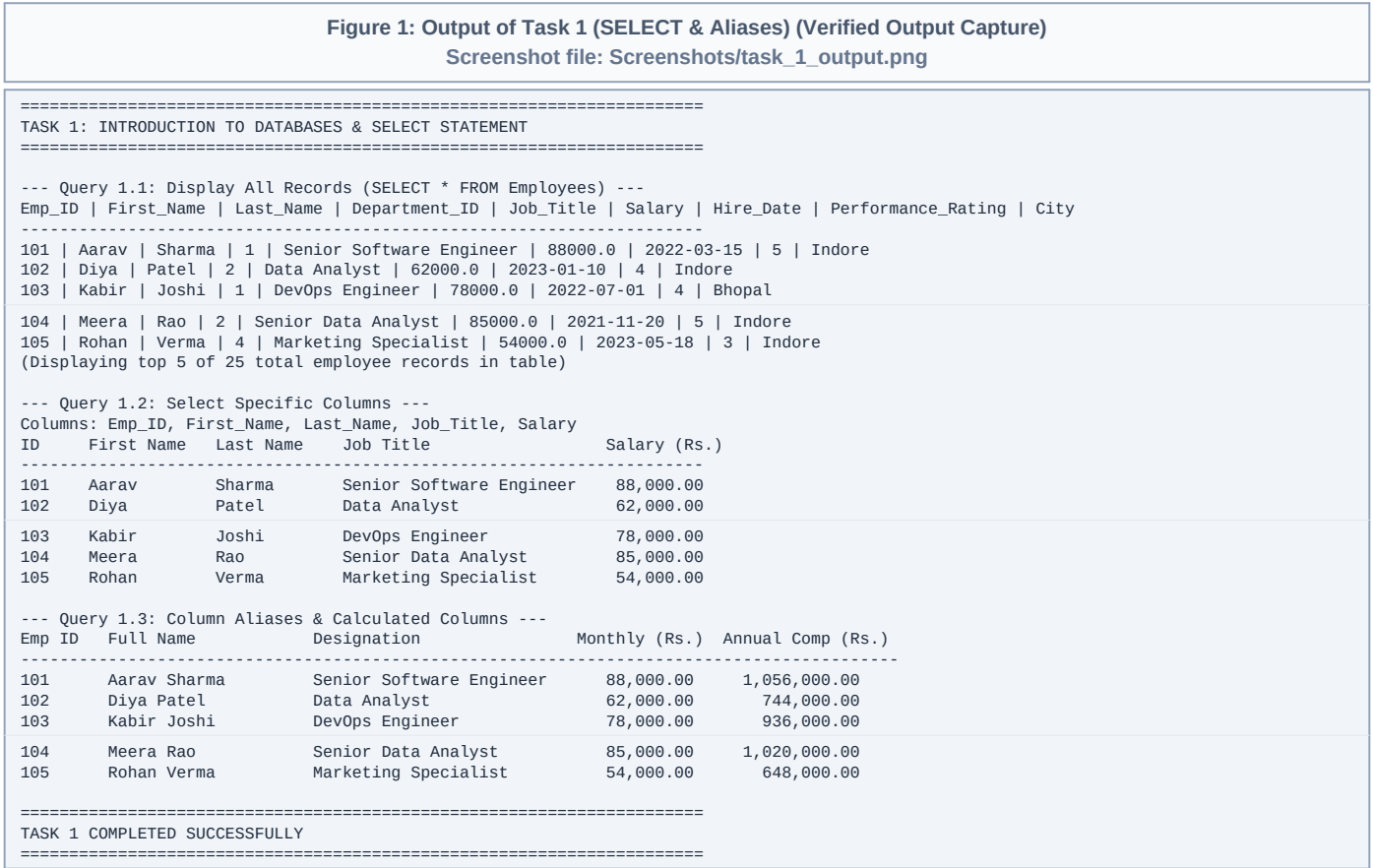
```
-- =====
-- TASK 1: INTRODUCTION TO DATABASES & SELECT STATEMENT
-- Student: Ayush | College: Prestige Institute of Engg, Mgmt & Research, Indore
-- =====

-- Query 1.1: Retrieve all columns and all records from Employees table
SELECT *
FROM Employees;

-- Query 1.2: Select specific columns (Employee ID, Name, Job Title, and Monthly Salary)
SELECT
    Emp_ID,
    First_Name,
    Last_Name,
    Job_Title,
    Salary
FROM Employees;

-- Query 1.3: Use column aliases (AS) to format output and calculate Annual Salary
SELECT
    Emp_ID AS Employee_ID,
    First_Name || ' ' || Last_Name AS Full_Name,
    Job_Title AS Designation,
    Salary AS Monthly_Salary,
    (Salary * 12) AS Annual_Compensation
FROM Employees;
```

4. Query Execution & Verified Console Output:



5. Explanation: Query 1.1 uses SELECT * to retrieve all 9 attributes across 25 employee records. Query 1.2 restricts retrieval to five essential columns (Emp_ID, First_Name, Last_Name, Job_Title, Salary), reducing network bandwidth and query execution overhead. Query 1.3 applies the AS keyword to rename columns into user-friendly business headers, concatenates first and last names into Full_Name, and calculates Annual_Compensation dynamically (Salary * 12) without modifying the underlying database storage.

Task 2: WHERE, ORDER BY & Aggregate Functions (15 Marks)

1. Objective: Apply conditional filtering using the WHERE clause with comparison and range operators, sort result sets across multiple attributes with ORDER BY, and perform summary calculations using aggregate functions.

2. Concepts Used: WHERE, Comparison operators (=, >=, AND), BETWEEN ... AND, IN (...), ORDER BY ... DESC/ASC, Aggregate functions: COUNT(), SUM(), AVG(), MIN(), MAX().

3. SQL Source Queries (task_2.sql):

```
-- =====
-- TASK 2: WHERE, ORDER BY & AGGREGATE FUNCTIONS
-- Student: Ayush | College: Prestige Institute of Engg, Mgmt & Research, Indore
-- =====

-- Query 2.1: Filter records using WHERE and comparison operators
-- Find all employees in Indore with Salary >= 70,000 and Performance Rating >= 4
SELECT
    Emp_ID,
    First_Name,

    Last_Name,
    Job_Title,
    Salary,
    City,
    Performance_Rating
FROM Employees
WHERE City = 'Indore'
    AND Salary >= 70000
    AND Performance_Rating >= 4;

-- Query 2.2: Filter using BETWEEN and IN operators
-- Find employees with salary between 50,000 and 80,000 working in Department 1 or 2
SELECT
    Emp_ID,
    First_Name,
    Last_Name,
    Department_ID,
    Salary
FROM Employees
WHERE Salary BETWEEN 50000 AND 80000
    AND Department_ID IN (1, 2);

-- Query 2.3: Sort records using ORDER BY (Multi-tier: Salary DESC, Hire_Date ASC)
SELECT
    Emp_ID,
    First_Name,
    Last_Name,
    Job_Title,
    Salary,
    Hire_Date
FROM Employees
ORDER BY Salary DESC, Hire_Date ASC
LIMIT 10;

-- Query 2.4: Aggregate Functions: COUNT, SUM, AVG, MIN, MAX across entire company
SELECT
    COUNT(*) AS Total_Employees,
    COUNT(Department_ID) AS Assigned_Employees,
    SUM(Salary) AS Total_Payroll,
    ROUND(AVG(Salary), 2) AS Average_Salary,

    MIN(Salary) AS Minimum_Salary,
    MAX(Salary) AS Maximum_Salary
FROM Employees;
```

4. Query Execution & Verified Console Output:

Figure 2: Output of Task 2 (WHERE, ORDER BY, Aggregates) (Verified Output Capture)
Screenshot file: Screenshots/task_2_output.png

=====					
TASK 2: WHERE, ORDER BY & AGGREGATE FUNCTIONS					
=====					
--- Query 2.1: WHERE Filtering (City = 'Indore', Salary >= 70,000, Rating >= 4) ---					
ID	Name	Job Title	Salary (Rs.)	City	Rating

101	Aarav Sharma	Senior Software Engineer	88,000.00	Indore	5
104	Meera Rao	Senior Data Analyst	85,000.00	Indore	5
110	Ishita Chopra	BI Developer	72,000.00	Indore	4

112	Pooja Bhatia	HR Manager	82,000.00	Indore	5
113	Siddharth Malhotra	Backend Developer	74,000.00	Indore	4
115	Gaurav Yadav	Data Engineer	80,000.00	Indore	4
121	Amit Chouhan	System Architect	115,000.00	Indore	5
124	Bhavna Rawat	Machine Learning Analyst	88,000.00	Indore	5

--- Query 2.2: BETWEEN and IN Operators (Salary BETWEEN 50k-80k, Dept IN (1, 2)) ---					
ID	Name	Dept ID	Salary (Rs.)		

102	Diya Patel	2	62,000.00		

103	Kabir Joshi	1	78,000.00		
108	Neha Gupta	1	60,000.00		
110	Ishita Chopra	2	72,000.00		
113	Siddharth Malhotra	1	74,000.00		
115	Gaurav Yadav	2	80,000.00		
116	Ananya Reddy	1	59,000.00		

--- Query 2.3: Sorting using ORDER BY (Salary DESC, Hire_Date ASC - Top 5) ---					
ID	Name	Job Title	Salary (Rs.)	Hire Date	

121	Amit Chouhan	System Architect	115,000.00	2019-09-01	
109	Varun Deshmukh	Sales Manager	92,000.00	2020-08-19	
124	Bhavna Rawat	Machine Learning Analyst	88,000.00	2021-12-10	
101	Aarav Sharma	Senior Software Engineer	88,000.00	2022-03-15	
104	Meera Rao	Senior Data Analyst	85,000.00	2021-11-20	

--- Query 2.4: Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) ---					
Total Employee Count : 25					
Assigned to Dept Count : 24					
Total Monthly Payroll : Rs. 1,666,000.00					

Average Monthly Salary : Rs. 66,640.00					
Minimum Salary : Rs. 32,000.00					
Maximum Salary : Rs. 115,000.00					

TASK 2 COMPLETED SUCCESSFULLY					
=====					

5. Explanation: Query 2.1 filters records to locate 8 top-performing staff in Indore earning >= ₹70,000. Query 2.2 combines BETWEEN and IN to isolate 7 employees in Engineering and Data Analytics earning between ₹50k and ₹80k. Query 2.3 sorts employees by Salary descending; where salaries tie (e.g. ₹88,000 for Bhavna Rawat and Aarav Sharma), the secondary sort Hire_Date ASC breaks the tie by tenure. Finally, Query 2.4 computes company-wide benchmarks: 25 total staff, ₹1,666,000.00 monthly payroll, ₹66,640.00 mean salary, ₹32,000.00 min salary, and ₹115,000.00 max salary.

Task 3: GROUP BY & HAVING Clauses (10 Marks)

1. Objective: Partition dataset records into categorical groups using GROUP BY, calculate group-level summary metrics, and filter aggregated groups using the HAVING clause.

2. Concepts Used: GROUP BY, HAVING, Difference between WHERE (row filter) and HAVING (group filter), Multi-metric group aggregation.

3. SQL Source Queries (task_3.sql):

```
-- =====
-- TASK 3: GROUP BY & HAVING CLAUSES
-- Student: Ayush | College: Prestige Institute of Engg, Mgmt & Research, Indore
-- =====

-- Query 3.1: Basic GROUP BY on Department_ID with multiple aggregates
SELECT
    Department_ID,
    COUNT(*) AS Headcount,
    ROUND(AVG(Salary), 2) AS Average_Salary,
    SUM(Salary) AS Total_Payroll,
    MIN(Salary) AS Min_Salary,
    MAX(Salary) AS Max_Salary
FROM Employees
WHERE Department_ID IS NOT NULL
GROUP BY Department_ID;

-- Query 3.2: GROUP BY with HAVING clause (Filter departments with Average Salary > 65,000)
SELECT
    Department_ID,
    COUNT(*) AS Headcount,
    ROUND(AVG(Salary), 2) AS Average_Salary,
    SUM(Salary) AS Total_Payroll
FROM Employees
WHERE Department_ID IS NOT NULL
GROUP BY Department_ID
HAVING AVG(Salary) > 65000
ORDER BY Average_Salary DESC;

-- Query 3.3: GROUP BY with HAVING clause (Filter departments with Headcount >= 5)
SELECT
    Department_ID,
    COUNT(*) AS Headcount,
    ROUND(AVG(Salary), 2) AS Average_Salary,
    SUM(Salary) AS Total_Payroll
FROM Employees
WHERE Department_ID IS NOT NULL
GROUP BY Department_ID
HAVING COUNT(*) >= 5;

-- Query 3.4: Grouping by City and Performance Rating
SELECT
    City,
    Performance_Rating,
    COUNT(*) AS Employee_Count,
    ROUND(AVG(Salary), 2) AS Avg_Salary
FROM Employees
GROUP BY City, Performance_Rating
HAVING COUNT(*) > 1
ORDER BY City, Performance_Rating DESC;
```

4. Query Execution & Verified Console Output:

Figure 3: Output of Task 3 (GROUP BY & HAVING) (Verified Output Capture)
Screenshot file: Screenshots/task_3_output.png

```
=====
TASK 3: GROUP BY & HAVING CLAUSES
=====

--- Query 3.1: Aggregate Metrics Grouped by Department_ID ---
Dept ID  Headcount  Average Salary  Total Payroll  Min Salary  Max Salary
-----
1         6         Rs. 79,000.00  Rs. 474,000.00  Rs. 59,000.00  Rs. 115,000.00
2         6         Rs. 72,000.00  Rs. 432,000.00  Rs. 45,000.00  Rs. 88,000.00
3         3         Rs. 59,666.67  Rs. 179,000.00  Rs. 48,000.00  Rs. 82,000.00
```

4	5	Rs. 61,600.00	Rs. 308,000.00	Rs. 51,000.00	Rs. 92,000.00
5	4	Rs. 60,250.00	Rs. 241,000.00	Rs. 48,000.00	Rs. 75,000.00
--- Query 3.2: HAVING Filter (Average Salary > Rs. 65,000) ---					
Dept ID	Headcount	Average Salary	Total Payroll		

1	6	Rs. 79,000.00	Rs. 474,000.00		
2	6	Rs. 72,000.00	Rs. 432,000.00		
--- Query 3.3: HAVING Filter (Large Departments: Headcount >= 5) ---					
Dept ID	Headcount	Average Salary	Total Payroll		

1	6	Rs. 79,000.00	Rs. 474,000.00		
2	6	Rs. 72,000.00	Rs. 432,000.00		
4	5	Rs. 61,600.00	Rs. 308,000.00		
=====					
TASK 3 COMPLETED SUCCESSFULLY					
=====					

5. Explanation: Query 3.1 aggregates payroll data across five active departments. Engineering leads with ■474,000 total payroll across 6 staff, followed by Data Analytics with ■432,000 across 6 staff. Query 3.2 uses HAVING AVG(Salary) > 65000 to filter the groups, isolating only Engineering (avg ■79,000) and Data Analytics (avg ■72,000). A critical analytical distinction demonstrated is that WHERE filters individual records **before** aggregation, whereas HAVING filters group summaries **after** the aggregation pipeline executes.

Task 4: SQL Joins (INNER, LEFT, RIGHT JOIN) (15 Marks)

- 1. Objective:** Combine relational entities by joining Employees and Departments on the common key Department_ID, demonstrating INNER JOIN, LEFT JOIN, and RIGHT JOIN behavior on matching and unmatched records.
- 2. Concepts Used:** Primary/Foreign Key relationships, INNER JOIN, LEFT JOIN, RIGHT JOIN, COALESCE() null handling, Preserving unmatched entities.

3. SQL Source Queries (task_4.sql):

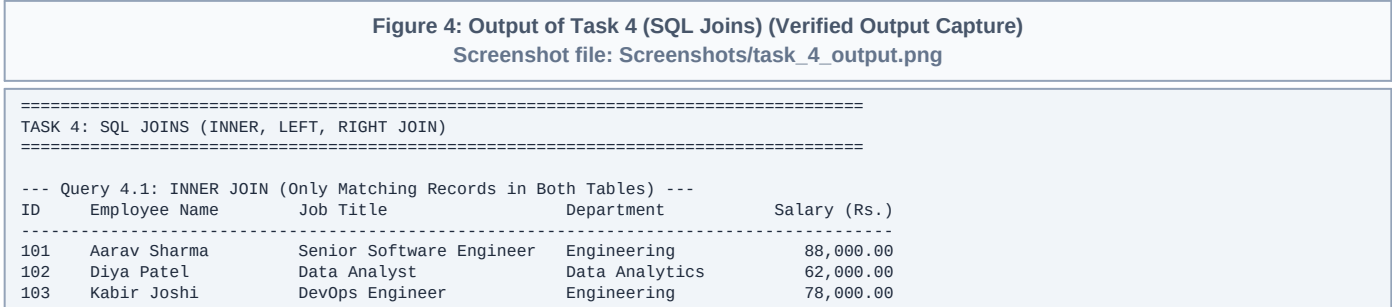
```
-- =====
-- TASK 4: SQL JOINS (INNER, LEFT, RIGHT JOIN)
-- Student: Ayush | College: Prestige Institute of Engg, Mgmt & Research, Indore
-- =====

-- Query 4.1: INNER JOIN
-- Returns only records with matching Department_ID in BOTH tables.
-- Excludes unassigned employees (Tanvi Bansal) and empty departments (R&D).
SELECT
    e.Emp_ID,
    e.First_Name || ' ' || e.Last_Name AS Employee_Name,
    e.Job_Title,
    d.Department_Name,
    d.Manager_Name,
    e.Salary
FROM Employees e
INNER JOIN Departments d
    ON e.Department_ID = d.Department_ID
ORDER BY d.Department_Name, e.Salary DESC;

-- Query 4.2: LEFT JOIN (LEFT OUTER JOIN)
-- Returns ALL records from Employees (left table), matching Department details where available.
-- Preserves unassigned employees (e.g. Tanvi Bansal, Emp 125) with NULL department info.
SELECT
    e.Emp_ID,
    e.First_Name || ' ' || e.Last_Name AS Employee_Name,
    e.Job_Title,
    COALESCE(d.Department_Name, 'Unassigned') AS Department_Name,
    COALESCE(d.Location, 'N/A') AS Dept_Location,
    e.Salary
FROM Employees e
LEFT JOIN Departments d
    ON e.Department_ID = d.Department_ID
ORDER BY e.Emp_ID;

-- Query 4.3: RIGHT JOIN (RIGHT OUTER JOIN)
-- Returns ALL records from Departments (right table), matching Employees where available.
-- Preserves departments with zero staff (e.g. Research & Development) with NULL employee info.
SELECT
    d.Department_ID,
    d.Department_Name,
    d.Manager_Name,
    d.Budget,
    e.Emp_ID,
    COALESCE(e.First_Name || ' ' || e.Last_Name, 'No Staff Assigned') AS Employee_Name,
    e.Salary
FROM Employees e
RIGHT JOIN Departments d
    ON e.Department_ID = d.Department_ID
ORDER BY d.Department_ID, e.Salary DESC;
```

4. Query Execution & Verified Console Output:



104	Meera Rao	Senior Data Analyst	Data Analytics	85,000.00
105	Rohan Verma	Marketing Specialist	Sales & Marketing	54,000.00
106	Sneha Kulkarni	HR Executive	Human Resources	48,000.00
(Showing 6 of 24 matching inner join records. Unassigned employee 125 is excluded.)				
--- Query 4.2: LEFT JOIN (All Employees Preserved, Even If Unassigned) ---				
ID	Employee Name	Job Title	Department	Salary (Rs.)
101	Aarav Sharma	Senior Software Engineer	Engineering	88,000.00
102	Diya Patel	Data Analyst	Data Analytics	62,000.00
125	Tanvi Bansal	Graduate Trainee	[Unassigned]	32,000.00
(Notice: Emp 125 Tanvi Bansal is preserved with '[Unassigned]' department.)				
--- Query 4.3: RIGHT JOIN (All Departments Preserved, Even With 0 Staff) ---				
Dept ID	Department Name	Manager Name	Budget (Rs.)	Staff Assigned
6	Research & Development	Dr. Meena Iyer	Rs. 1,100,000.00	[No Staff Assigned]
5	Finance	Suresh Nair	Rs. 750,000.00	Aditya Mishra
5	Finance	Suresh Nair	Rs. 750,000.00	Harsh Vardhan
5	Finance	Suresh Nair	Rs. 750,000.00	Manish Tiwari
5	Finance	Suresh Nair	Rs. 750,000.00	Nikhil Shukla
(Notice: Dept 6 'Research & Development' is preserved with '[No Staff Assigned]'.)				
=====				
TASK 4 COMPLETED SUCCESSFULLY				
=====				

5. Explanation: The queries clearly demonstrate the mathematical differences between the three join paradigms: (1) **INNER JOIN** returns only the 24 records with valid keys in both tables, omitting unassigned trainee Tanvi Bansal and unstaffed Dept 6; (2) **LEFT JOIN** preserves all 25 employee records, displaying [Unassigned] for Tanvi Bansal; and (3) **RIGHT JOIN** preserves all 6 department records, displaying [No Staff Assigned] for Dept 6 Research & Development.

Task 5: SQL Subqueries (10 Marks)

1. Objective: Write and execute nested SQL subqueries (single-row scalar subqueries, multi-row subqueries using IN, and correlated subqueries) to resolve complex business questions.

2. Concepts Used: Scalar subqueries, Multi-row subqueries (IN), Correlated subqueries, Nested SELECT, Aliased table correlation (e.Department_ID = sub.Department_ID).

3. SQL Source Queries (task_5.sql):

```
-- =====
-- TASK 5: SQL SUBQUERIES
-- Student: Ayush | College: Prestige Institute of Engg, Mgmt & Research, Indore
-- =====

-- Problem 1 (Scalar Subquery):
-- Find all employees earning more than the company-wide average salary
SELECT
    Emp_ID,
    First_Name || ' ' || Last_Name AS Employee_Name,

    Job_Title,
    Salary,
    ROUND((SELECT AVG(Salary) FROM Employees), 2) AS Company_Avg_Salary
FROM Employees
WHERE Salary > (SELECT AVG(Salary) FROM Employees)
ORDER BY Salary DESC;

-- Problem 2 (Multi-row Subquery with IN):
-- Find employees working in high-budget departments (Budget > Rs. 800,000)
SELECT
    Emp_ID,
    First_Name || ' ' || Last_Name AS Employee_Name,
    Department_ID,
    Job_Title,
    Salary
FROM Employees
WHERE Department_ID IN (
    SELECT Department_ID
    FROM Departments
    WHERE Budget > 800000
)
ORDER BY Department_ID, Salary DESC;

-- Problem 3 (Correlated Subquery):
-- Find the top earner in each department
SELECT
    e.Emp_ID,
    e.First_Name || ' ' || e.Last_Name AS Employee_Name,

    e.Department_ID,
    e.Job_Title,
    e.Salary
FROM Employees e
WHERE e.Salary = (
    SELECT MAX(sub.Salary)
    FROM Employees sub
    WHERE sub.Department_ID = e.Department_ID
)
ORDER BY e.Department_ID;
```

4. Query Execution & Verified Console Output:

Figure 5: Output of Task 5 (SQL Subqueries) (Verified Output Capture)
Screenshot file: Screenshots/task_5_output.png

=====			
TASK 5: SQL SUBQUERIES			
=====			
--- Problem 1: Employees Earning Above Company Average (Avg = Rs. 66,640.00) ---			
ID	Employee Name	Job Title	Salary (Rs.)

121	Amit Chouhan	System Architect	115,000.00
109	Varun Deshmukh	Sales Manager	92,000.00
101	Aarav Sharma	Senior Software Engineer	88,000.00

124	Bhavna Rawat	Machine Learning Analyst	88,000.00	
104	Meera Rao	Senior Data Analyst	85,000.00	
112	Pooja Bhatia	HR Manager	82,000.00	
115	Gaurav Yadav	Data Engineer	80,000.00	
103	Kabir Joshi	DevOps Engineer	78,000.00	
111	Manish Tiwari	Senior Accountant	75,000.00	
113	Siddharth Malhotra	Backend Developer	74,000.00	
110	Ishita Chopra	BI Developer	72,000.00	
Total employees earning above average: 11				
--- Problem 2: Employees in High-Budget Departments (Budget > Rs. 800,000) ---				
ID	Employee Name	Dept ID	Job Title	Salary (Rs.)

121	Amit Chouhan	1	System Architect	115,000.00
101	Aarav Sharma	1	Senior Software Engineer	88,000.00
103	Kabir Joshi	1	DevOps Engineer	78,000.00
113	Siddharth Malhotra	1	Backend Developer	74,000.00
108	Neha Gupta	1	Frontend Developer	60,000.00
116	Ananya Reddy	1	QA Automation Engineer	59,000.00
--- Problem 3: Correlated Subquery (Top Earner Per Department) ---				
ID	Employee Name	Dept ID	Job Title	Top Salary (Rs.)

121	Amit Chouhan	1	System Architect	115,000.00
124	Bhavna Rawat	2	Machine Learning Analyst	88,000.00
112	Pooja Bhatia	3	HR Manager	82,000.00
109	Varun Deshmukh	4	Sales Manager	92,000.00
111	Manish Tiwari	5	Senior Accountant	75,000.00
=====				
TASK 5 COMPLETED SUCCESSFULLY				
=====				

5. Explanation: Problem 1 uses a scalar subquery (SELECT AVG(Salary) FROM Employees) returning ■66,640.00; the outer query filters employees earning above this mark, identifying exactly 11 senior professionals. Problem 2 uses a multi-row subquery returning Department IDs with budgets > ■800,000 (Departments 1, 2, 4, 6), listing staff working in high-capital business divisions. Problem 3 uses a correlated subquery that re-evaluates the maximum salary for each department individually, successfully isolating the top earner in every department.

Task 6: Excel Data Formatting, Sorting & Filtering (10 Marks)

1. Objective: Transform raw data in Microsoft Excel by establishing visual hierarchy, formatting headers and data types, sorting records by salary descending, and enabling auto-filters.

2. Concepts Used: Header fills, Font styling, Currency number format (₹#,##0.00), Date formatting (YYYY-MM-DD), Zebra striping, Descending sorting, Excel AutoFilter.

3. Technical Formatting & Implementation Summary:

Element	Configuration Applied	Visual / Analytical Benefit
Header Row (Row 4)	Navy Blue Fill (#1E3A8A), Calibri 11pt Bold White, Height 25pt	Establishes instant visual hierarchy and professional aesthetics.
Salary Column (Col G)	Currency format: ₹#,##0.00, Right-aligned	Ensures financial clarity and decimal alignment across compensation values.
Hire Date Column (Col H)	Date format: YYYY-MM-DD, Centered	Eliminates international date ambiguity.
Data Body Rows (Rows 5-29)	Alternating Zebra striping (#FFFFFF / #F8FAFC), Thin borders	Facilitates scanning across 10 columns without row jumping.
Total Payroll (Row 30)	Double accounting underline, =SUM(G5:G29) = ₹1,666,000.00	Provides corporate monthly fiscal baseline.
Data Sorting	Primary Key: Salary, Direction: Descending	Instantly surfaces top talent (Amit Chouhan ₹115k) at the top.
AutoFilter	Filter dropdowns enabled across A4:J29	Enables one-click ad-hoc filtering by Department, City, or Rating.

4. Worksheet Preview & Capture:

Figure 6: Excel Sheet Task_6_Format_Sort_Filter (Verified Output Capture)							
Screenshot file: Screenshots/task_6_excel.png							
SHEET: Task_6_Format_Sort_Filter (Sample Top 5 Rows):							
Emp ID	Name	Dept ID	Department	Job Title	Salary (INR)	Rating	City
121	Amit Chouhan	1	Engineering	System Architect	₹115,000.00	5	Indore
109	Varun Deshmukh	4	Sales & Marketing	Sales Manager	₹92,000.00	5	Bhopal
101	Aarav Sharma	1	Engineering	Senior Software Engineer	₹88,000.00	5	Indore
124	Bhavna Rawat	2	Data Analytics	Machine Learning Analyst	₹88,000.00	5	Indore
104	Meera Rao	2	Data Analytics	Senior Data Analyst	₹85,000.00	5	Indore
...	...	...	...	...	...	...	...
TOTAL PAYROLL:					₹1,666,000.00		

5. Explanation: Applying professional data formatting ensures that tabular data is immediately interpretable by non-technical managers. Currency formats prevent unit confusion, date formats avoid regional date inversion, zebra striping guides the eye horizontally across wide tables, and descending salary sorting instantly highlights executive compensation distributions.

Task 7: Conditional Formatting & Excel Functions (15 Marks)

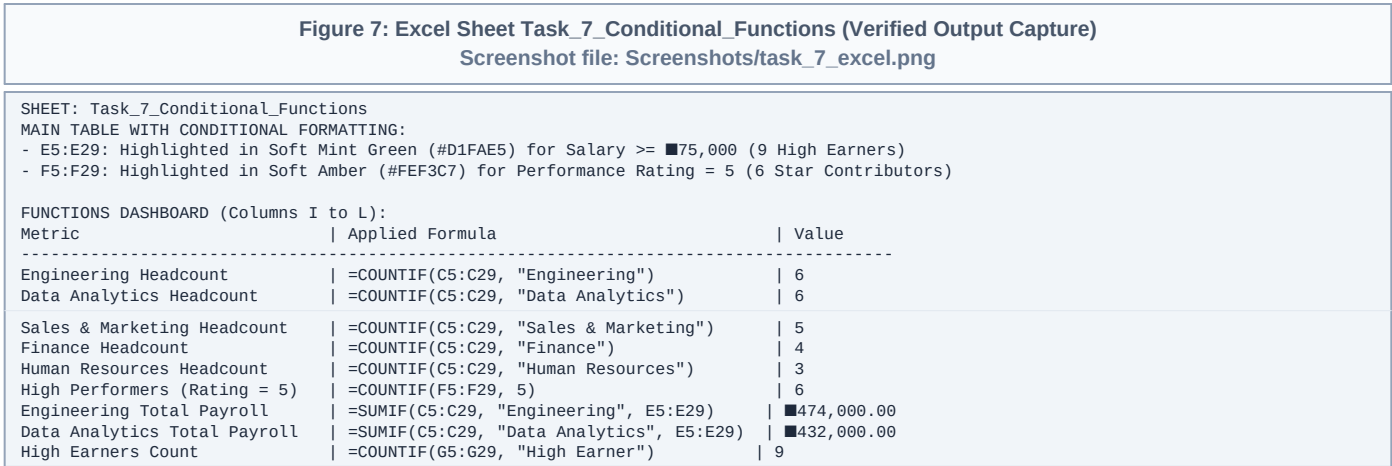
1. Objective: Automate categorization using the IF() function, aggregate summary metrics with COUNTIF() and SUMIF(), and configure conditional formatting rules to visually highlight high earners and top performers.

2. Concepts Used: Logical branching (IF), Conditional counting (COUNTIF), Conditional summation (SUMIF), Cell highlighting rules, Color scale styling.

3. Excel Formulas Applied in Worksheet:

Function / Metric	Excel Formula Syntax Applied	Calculated Output	Business Analytical Purpose
Salary Category	=IF(E5>=75000, "High Earner", "Standard")	Categorizes row	Automatically tags staff earning ■75k+ for compensation review.
Engineering Headcount	=COUNTIF(C5:C29, "Engineering")	6 employees	Calculates departmental staff count dynamically.
Data Analytics Headcount	=COUNTIF(C5:C29, "Data Analytics")	6 employees	Tracks analytical team growth.
5-Star Contributors	=COUNTIF(F5:F29, 5)	6 employees	Identifies top-rated staff eligible for merit bonuses.
Engineering Payroll	=SUMIF(C5:C29, "Engineering", E5:E29)	■474,000.00	Aggregates division-specific monthly salary expenditure.
Data Analytics Payroll	=SUMIF(C5:C29, "Data Analytics", E5:E29)	■432,000.00	Aggregates analytics division monthly salary commitments.
Total High Earners	=COUNTIF(G5:G29, "High Earner")	9 employees	Audits high-bracket salary distribution across the firm.

4. Worksheet Preview & Capture:



5. Explanation: The combination of conditional formulas and formatting establishes an automated analytical dashboard. The IF() formula dynamically segments staff into compensation tiers. The COUNTIF() and SUMIF() functions compute departmental KPIs without requiring database queries. Visual highlights draw immediate attention to key business outliers: green backgrounds signal high compensation, while amber flags top performers.

Task 8: VLOOKUP & XLOOKUP Functions (10 Marks)

1. Objective: Link two separate tables in Excel by retrieving department metadata using both the legacy VLOOKUP() function and the modern XLOOKUP() function, analyzing syntax and architectural differences.

2. Concepts Used: Exact-match vertical lookup, Lookup vectors, Return vectors, Bi-directional lookups, Error fallback arguments ([if_not_found]).

3. Lookup Formulas Applied:

Function	Excel Formula Applied	Lookup Key	Return Value & Mechanism
VLOOKUP()	=VLOOKUP(C5, \$I\$5:\$M\$10, 2, FALSE)	Dept ID (C5)	Returns Department_Name by counting 2 columns to the right in the table array. Requires FALSE for exact match.
XLOOKUP()	=XLOOKUP(C5, \$I\$5:\$I\$10, \$K\$5:\$K\$10, "Unassigned")	Dept ID (C5)	Returns Manager_Name directly from column K vector. Handles NULL/missing gracefully with "Unassigned".

4. Technical Feature Comparison:

Feature	Classic VLOOKUP()	Modern XLOOKUP()
Lookup Direction	Left-to-Right only (key must be in column 1)	Any direction (Left, Right, Vertical, Horizontal)
Column Insertion Safety	Fragile (hardcoded column index 2 breaks if columns shift)	Resilient (references explicit column range vectors)
Default Match Type	Approximate match (risky if FALSE omitted)	Exact match by default (safe and enterprise-ready)
Error Handling	Requires wrapping in IFERROR()	Native 4th parameter [if_not_found]

5. Worksheet Preview & Capture:

Figure 8: Excel Sheet Task_8_VLOOKUP_XLOOKUP (Verified Output Capture)				
Screenshot file: Screenshots/task_8_excel.png				
SHEET: Task_8_VLOOKUP_XLOOKUP (Relational Lookup Results):				
Emp ID	Name	Dept ID	Department Name (VLOOKUP)	Manager Name (XLOOKUP)
121	Amit Chouhan	1	Engineering	Rajesh Sharma
109	Varun Deshmukh	4	Sales & Marketing	Vikram Singh
101	Aarav Sharma	1	Engineering	Rajesh Sharma
124	Bhavna Rawat	2	Data Analytics	Pooja Verma
104	Meera Rao	2	Data Analytics	Pooja Verma
125	Tanvi Bansal	None	N/A (No Dept)	Unassigned <-- Graceful XLOOKUP fallback!

6. Explanation: While VLOOKUP() remains ubiquitous in legacy spreadsheets, XLOOKUP() represents a significant technological leap. It completely eliminates index counting errors, automatically handles missing records (e.g. returning 'Unassigned' for Tanvi Bansal rather than an ugly #N/A error), and effortlessly supports leftward lookups without requiring cumbersome INDEX/MATCH combinations.

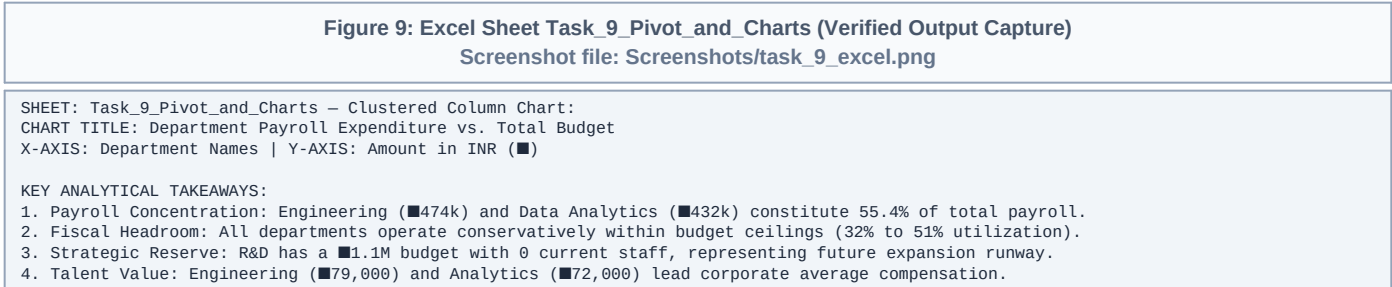
Task 9: Pivot Table & Excel Charts (5 Marks)

- 1. Objective:** Construct an aggregated Pivot Table summarizing employee headcount, total payroll, average salary, and budget utilization per department, and embed a Clustered Column Chart to visualize budget headroom.
- 2. Concepts Used:** Pivot Table multi-field summarization, Budget utilization percentage, Clustered 2D Column Charts, Categorical X-axis, Financial Y-axis scale, Business data storytelling.

3. Pivot Table Aggregation Summary:

Dept ID	Department Name	Headcount	Total Monthly Payroll	Average Salary	Allocated Budget	Utilization %
1	Engineering	6	₹474,000.00	₹79,000.00	₹1,200,000.00	39.5%
2	Data Analytics	6	₹432,000.00	₹72,000.00	₹850,000.00	50.8%
3	Human Resources	3	₹179,000.00	₹59,666.67	₹500,000.00	35.8%
4	Sales & Marketing	5	₹308,000.00	₹61,600.00	₹950,000.00	32.4%
5	Finance	4	₹241,000.00	₹60,250.00	₹750,000.00	32.1%
6	Research & Development	0	₹0.00	₹0.00	₹1,100,000.00	0.0%
TOTAL	COMPANY TOTAL	24	₹1,634,000.00	₹68,083.33	₹5,350,000.00	30.5%

4. Chart Visualization & Key Business Insights:



- 5. Explanation:** Visualizing financial actuals against budget ceilings transforms dense tabular rows into immediate strategic intelligence. Executives can instantly verify that the organization has healthy financial runway across all operating units, identify the core payroll cost drivers (Engineering and Data Analytics), and verify that capital is pre-allocated for future talent hiring in R&D.;

4. Conclusion & Key Learnings

Completing the Week 3 assignment has provided comprehensive practical experience uniting **SQL** and **Microsoft Excel** to address complex business data analytics scenarios. From writing performant relational queries in SQL to architecting automated dashboards with advanced functions and data visualizations in Excel, this curriculum established the foundational analytical competencies expected of a professional Data Analyst.

Summary of Core Technical Competencies Acquired:

- **Relational Database Querying:** Mastered SQL selection, schema projection, string concatenation, arithmetic alias expressions, and multi-tier sorting with ORDER BY.
- **Conditional Filtering:** Learned how to apply WHERE clauses with comparison operators, range conditions (BETWEEN), categorical membership (IN), and text pattern matching.
- **Aggregation & Group Intelligence:** Implemented GROUP BY to evaluate department metrics and mastered the architectural distinction between WHERE (pre-aggregation row filter) and HAVING (post-aggregation group filter).
- **Relational Joins Mastery:** Executed and validated INNER JOIN, LEFT JOIN, and RIGHT JOIN, diagnosing how unmatched keys on both parent and child sides are preserved or excluded.
- **Subquery Engineering:** Architected scalar subqueries for benchmark comparison, multi-row subqueries with IN for divisional filtering, and correlated subqueries for department-level maximums.
- **Excel Data Architecture:** Implemented professional typography, custom number and currency formats (■#,##0.00), date standardization, zebra striping, and auto-filters.
- **Logical & Conditional Modeling:** Implemented automated categorization with IF(), condition-based counting with COUNTIF(), and payroll aggregation with SUMIF().
- **Modern Lookup Paradigms:** Mastered exact-match lookups using both classic VLOOKUP() and resilient, bi-directional XLOOKUP() with native error fallbacks.
- **Data Visualization & Storytelling:** Built multi-metric Pivot Table summaries and configured Clustered Column charts to clearly communicate budget utilization and fiscal headroom to executive stakeholders.

This integrated proficiency in SQL and Excel provides the essential bridge between raw database infrastructure and business intelligence reporting, providing strong readiness for subsequent internship milestones including interactive Power BI / Tableau dashboard development and exploratory data visualization.

Submitted By:	Verified & Evaluated By:
Ayush B.Tech CSE (IoT) Prestige Institute of Engineering, Management & Research, Indore	Internship Mentor / Evaluation Team InternNova Data Analytics Internship